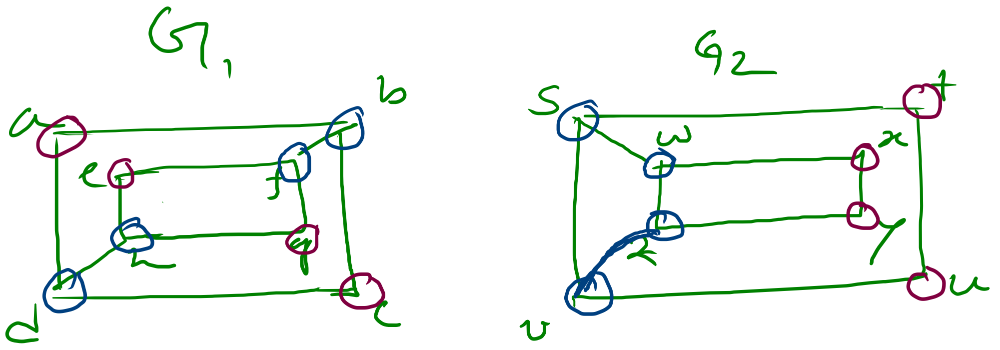
Isomorphism

(*) Same no. of elements despite different structure.

(*) A property preserved by isomorphism is called graph invariant.

(*) How to identify?

1. Same no. of vertices
2. " " " edges
3. " " " vertices with same degree.
4. A function.



	G_1	G_2
# of V	8	8
# of E	10	10
# of V with same deg	$d_2 \to 4$ $d_3 \to 4$	$d_2 \to 4$ $d_3 \to 4$

⊛ derive an isomorphic function.

$$A_{G_1} = \begin{array}{c|cccccccc} & a & b & c & d & e & f & g & h \\ \hline a & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ b & 1 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ c & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ d & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 1 \\ e & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ f & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\ g & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 1 \\ h & 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 \end{array}$$

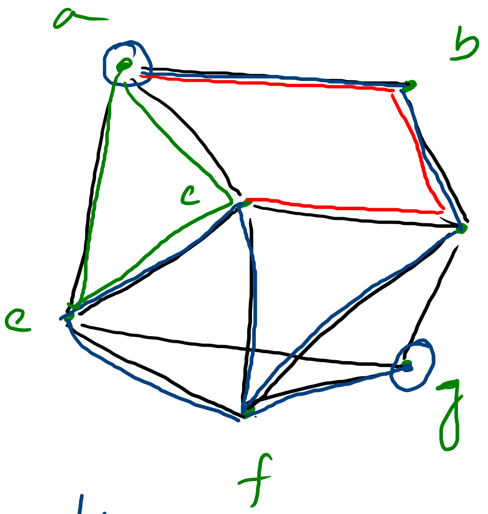
$$A_{G_2} = \begin{array}{c|cccccccc} & s & t & u & v & w & x & y & z \\ \hline s & 0 & 1 & 0 & 1 & 1 & 0 & 0 & 0 \\ t & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ a \rightarrow u & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ v & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 1 \\ w & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ x & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\ y & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ z & 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 \end{array}$$

Since A_{G_1}, A_{G_2} are not preservable

So, they are not isomorphic.

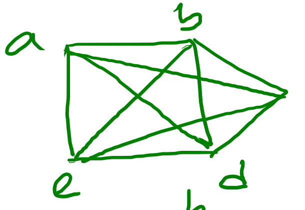
10.4. Connectivity

* Path → #edges → path length = n
* Circuit
* Simple path → a path with no repeating edges.



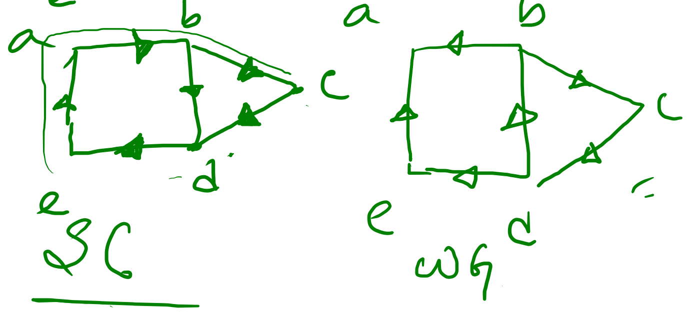
path
simple path
circuit.

* Connected graph → every pair of distinct vertices has an edge in UG.



→ Connected

* DG
  * → strongly connected → path(a,b) or path(b,a) where a,b ∈ V
  * → Weakly →



10.5 Euler and Hamiltonian path/circuit.

(*) Euler → all edges exactly once

- path → A simple path with every edge of G
- circuit → A " " " " " " G.
  - → every vertex must have even degree.

(*) Hamiltonian path → all vertices exactly once.

(*) HC
A graph with vertex of degree 1 has no HC.

EH → both must cover all vertices.
but
E → no repeating edges.
H → " " vertices